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Cooperative Learning of Shared Latent Dynamics for Distributed Online Optimization

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Abstract

We study networked agents that must agree on a decision minimizing the sum of their local costs, where each cost contains unmeasurable stochastic parameters that evolve with linear but unknown dynamics shared by all agents. Each agent receives only finitely many noisy gradient measurements and must then keep tracking the moving minimizer. Our scheme reconstructs the parameters from windows of gradients, learns the shared dynamics cooperatively by averaging instrumental-variable moment matrices over the network, and tracks the minimizer by gradient tracking on forecast gradients. We bound the error of every agent in learning the dynamics and in tracking the minimizer, and show that the learning error, and through it the tracking error, decreases with the number of agents. We demonstrate the approach on several examples.

Keywords: non-stationary online learning, distributed optimization, multi-task system identification, instrumental variables, gradient tracking, finite-sample analysis

1. Introduction

Online optimization is a fundamental tool for real-time decision making, with applications ranging from the control of robots to the recommendation of content, in which the cost to be minimized changes over time and its minimizer must be tracked as it moves (e.g., see Hazan, 2016; Simonetto et al., 2020; Dall'Anese et al., 2020; Wensing et al., 2024; Koren, 2010). In many multi-agent problems the cost changes through latent parameters: a group of agents must agree on a common decision, the cost of each agent depends on a private parameter that it cannot measure and whose dynamics it does not know, and each agent observes its cost only through noisy gradients, and only for a while. While existing methods track a moving minimizer with guarantees when the model of the parameters is known or gradients are available at every time, and learn unknown dynamics with guarantees when the states are measured, they offer no guarantees when the parameters are latent, the data are finite, and the information is dispersed and private, which is the typical situation in networked systems. This poses critical limitations on the deployment of online optimization for multi-agent decision making.

In this paper we propose a cooperative scheme for this problem and bounds that characterize its performance. In particular, we exploit that the agents are alike: their parameters follow the same unknown linear dynamics, so that the network can learn these dynamics as a single learner with all the data. Each agent reconstructs its parameters from windows of noisy gradients, the agents learn the dynamics by averaging instrumental-variable (IV) moment matrices over the network, which exchanges no data and avoids the bias that

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least squares suffers when regressor and residual share noise, and each agent tracks the minimizer by gradient tracking on the gradients forecast from the learned dynamics. Then, we derive bounds on the error of every agent in learning the dynamics and in tracking the minimizer. Our bounds are probabilistic in nature, and they depend explicitly on the number of agents, the size of the training data, the number of consensus rounds, and the window length: the agents learn the dynamics as if their data were pooled, this gain survives latent observations, windowing bias, and finitely many consensus rounds, and it carries over to the tracking error, whose remaining part is the irreducible error of forecasting a stochastic parameter. Finally, we illustrate our findings on two case studies and on a simulation in Webots with one process per agent.

Related work. Online learning in non-stationary environments tracks the minimizer of a time-varying cost up to a constant error (Besbes et al., 2015; Mokhtari et al., 2016; Hazan, 2016; Shalev-Shwartz, 2012); time-varying optimization improves on this with prediction–correction (Simonetto et al., 2016; Simonetto and Dall'Anese, 2017; Fazlyab et al., 2018), contracting dynamics (Davydov et al., 2025), or the internal model principle (Bianchin and Van Scoy, 2025, 2026; Bastianello et al., 2024; Casti et al., 2025), which embeds a known model of the parameter dynamics. Our prior work learned the model for a single agent (Tripathi et al., 2025, 2026). Distributed optimization converges linearly with constant step sizes by gradient tracking (Nedić and Ozdaglar, 2009; Xu et al., 2015; Di Lorenzo and Scutari, 2016; Nedić et al., 2017; Qu and Li, 2018), and its online versions track a moving minimizer up to its drift (Simonetto et al., 2017; Shahrampour and Jadbabaie, 2018; Carnevale et al., 2023; Notarnicola et al., 2023), but they need the local gradients at every time. In contrast, our agents receive only finitely many gradients, and at different locations. Non-asymptotic system identification (Simchowitz et al., 2018; Oymak and Ozay, 2019; Sarkar and Rakhlin, 2019; Tsiamis and Pappas, 2019; Ziemann and Tu, 2022; Ziemann et al., 2023) and its multi-task, federated, and meta-learning extensions (Wang et al., 2023; Toso et al., 2023; Modi et al., 2024; Toso et al., 2024) show that pooling data from similar systems reduces the sample complexity, but assume measured states, whereas ours are observed through noisy gradients. In contrast to federated learning (McMahan et al., 2017), which aggregates at a server, and to consensus-based fusion (Xiao et al., 2005), which estimates a static quantity, our agents learn the dynamics by exchanging only messages with their neighbors, and the data remain private.

Contributions.

- A three-layer scheme (Algorithm 1): windowed Gauss–Markov reconstruction of the parameters, cooperative identification of the dynamics by average consensus on IV moment matrices, and gradient tracking on the forecast gradients. The agents exchange only iterates and $p \times p$ matrices, where p is the parameter dimension, and never a measurement, a parameter, or a forecast.

- Theorem 2: with m agents and N gradients each, every agent identifies the dynamics matrix A with a statistical error of order $\sqrt{\log(m/\delta)/(mN)}$, as if one agent had all mN samples, plus a consensus term that decays geometrically in the number of rounds and a windowing bias set by the window length. The key concentration step (Lemma S1 in the

supplementary material) uses the Hanson–Wright inequality and needs no mixing-time argument.

- Theorem 4: the expected tracking error of every agent is a geometrically vanishing network transient plus a forecast error that inherits the $\sqrt{m}$ gain and converges to the irreducible error of forecasting a stochastic parameter.
- Two case studies and a Webots simulation with one process per agent confirm the $1/\sqrt{m}$ law and show that cooperation matters most when data are scarce; code for every figure and the simulation accompanies the paper.

2. Problem setup

Consider m agents that communicate over an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with $\mathcal{V} = \{1, \dots, m\}$, and with neighborhoods $\mathcal{N}_i$, and cooperatively solve the time-varying problem:

$$\min_{x \in \mathbb{R}^n} \; F(x, \theta(t)) := \sum_{i=1}^{m} f_i(x, \theta_i(t)), \qquad f_i(x, \theta_i) := f_i^0(x) + g_i(x)^{\mathsf{T}} \theta_i, \tag{1}$$

where $f_i^0 : \mathbb{R}^n \to \mathbb{R}$ and $g_i : \mathbb{R}^n \to \mathbb{R}^p$ are known and twice continuously differentiable, $\theta_i(t) \in \mathbb{R}^p$ is a latent parameter private to agent i, and $\theta(t) := [\theta_1(t)^{\mathsf{T}}, \dots, \theta_m(t)^{\mathsf{T}}]^{\mathsf{T}} \in \mathbb{R}^{mp}$. The parameters evolve under unknown linear stochastic dynamics common to all agents as:

$$\theta_i(t+1) = A\theta_i(t) + w_i(t), \qquad w_i(t) \sim \mathcal{N}(0, Q_i), \qquad i \in \mathcal{V}, \tag{2}$$

with unknown $A \in \mathbb{R}^{p \times p}$ and $Q_i \succeq 0$. Agent i has access only to a local gradient oracle:

$$y_i(x, t) = \nabla_x f_i^0(x) + C_i(x)\theta_i(t) + w_{m,i}(t) \tag{3}$$

with $w_{m,i}(t) \sim \mathcal{N}(0, R_i)$, $R_i \succ 0$, and $C_i(x) := \partial g_i^{\mathsf{T}} / \partial x \in \mathbb{R}^{n \times p}$ known; $\theta_i(0) \sim \mathcal{N}(0, P_{i,0})$, $w_i(t)$, and $w_{m,i}(t)$ are mutually independent across agents and time. Since $\nabla_x f_i^0(x)$ is known, $\tilde{y}_i(x, t) := y_i(x, t) - \nabla_x f_i^0(x) = C_i(x)\theta_i(t) + w_{m,i}(t)$.

Data collection: We collect the data for $t \in \{0, \dots, N-1\}$, where agent i queries its oracle at designed exploration points $x_i^{\mathrm{e}}(t)$ and stores $\tilde{y}_i(t) := \tilde{y}_i(x_i^{\mathrm{e}}(t), t)$ and $C_{i,t} := C_i(x_i^{\mathrm{e}}(t))$. During the *tracking phase* $t \geq N$ the oracle is no longer available and every agent produces an iterate $x_i(t)$.

Agent i knows only f_i^0, g_i, R_i and its own data, and exchanges one message with its neighbors at every time. The goal is to bound the expected tracking error $\mathbb{E}\|x_i(t) - x^*(t)\|$ of every agent for $t \geq N$, where $x^*(t)$ minimizes (1), and show that it that decreases with N and m.

Assumption 1 (Regularity and strong convexity) *There exist $0 < \mu \leq L$ and $\theta_i(t) \in \Theta_i \subseteq \mathbb{R}^p$, $\Theta := \Theta_1 \times \cdots \times \Theta_m$, such that $\nabla_x^2 F(x, \theta) \succeq \mu I_n$ and $\|\nabla_x^2 f_i(x, \theta_i)\| \leq L$ for all $x \in \mathbb{R}^n$, $\theta \in \Theta$, $i \in \mathcal{V}$.*

Assumption 2 (Stable shared dynamics) *A is Schur stable and invertible, $(A, Q_i^{1/2})$ is controllable.*

Assumption 3 (Persistent exploration) *The exploration points lie in a bounded set, and there exist $k_0 \geq \lceil p/n \rceil$ and $\alpha_0 > 0$ with $\sum_{j=0}^{k_0-1} C_{i,t+j}^{\mathsf{T}} C_{i,t+j} \succeq \alpha_0 I_p$ for all $i \in \mathcal{V}$ and $t \in \{0, \dots, N - k_0\}$.*

Assumption 4 (Communication) *$\mathcal{G}$ is connected, and $W \in \mathbb{R}^{m\times m}$ is symmetric and doubly stochastic with $W_{ij} > 0$ if and only if $j \in \mathcal{N}_i \cup \{i\}$.*

Assumption 1 makes $F(\cdot,\theta)$ μ-strongly convex for every $\theta \in \Theta$, with a unique minimizer $x^*(\theta)$; since $\nabla_x F(x^*(\theta),\theta') = C(x^*(\theta))(\theta' - \theta)$ with $C(x) := [C_1(x)\cdots C_m(x)]$, strong monotonicity gives for $\theta,\theta' \in \Theta$:

$$\|x^*(\theta') - x^*(\theta)\| \le L_*\|\theta' - \theta\|, \qquad L_* := \mu^{-1}\sup_{\theta\in\Theta}\|C(x^*(\theta))\|. \tag{4}$$

Assumption 2 yields $C_\rho \ge 1$ and $\rho_A \in (\rho(A), 1)$ with $\|A^j\| \le C_\rho \rho_A^j$, the stationary covariance $\Sigma_{\theta,i} := \sum_{l\ge 0} A^l Q_i (A^l)^\mathsf{T} \succ 0$, and $P_i(t) := \mathbb{E}[\theta_i(t)\theta_i(t)^\mathsf{T}]$ with $B_{\theta,i} := \sup_t(\operatorname{tr} P_i(t))^{1/2} < \infty$. Assumption 3 is a persistence-of-excitation condition on the exploration sequence. We define $\sigma := \|W - \mathbf{1}_m\mathbf{1}_m^\mathsf{T}/m\| < 1$.

3. Method: cooperative identification and distributed tracking

The proposed approach has three layers. First, a local reconstruction layer, which is the Gauss-Markov window estimator of the latent time-varying parameter. Second, a cooperative identification layer, which averages the instrumental-variable moments of all agents over the networks. Third, a network optimization layer, which is a gradient-tracking iteration driven by the oracle during data collection and by the local forecasts afterwards.

Layer 1: Local reconstruction of $\theta(t)$: For $t < N$, for a window length $k \ge k_0$, agent i stacks the measurements of each window, $\bar{y}_{i,t} := [\tilde{y}_i(t)^\mathsf{T},\dots,\tilde{y}_i(t+k-1)^\mathsf{T}]^\mathsf{T}$, $\bar{C}_{i,t} := [C_{i,t}^\mathsf{T},\dots,C_{i,t+k-1}^\mathsf{T}]^\mathsf{T}$, $\bar{R}_i := I_k \otimes R_i$, and computes the window estimates using **?** as

$$\tilde{\theta}_i(t) := K_{i,t}\bar{y}_{i,t}, \qquad K_{i,t} := (\bar{C}_{i,t}^\mathsf{T}\bar{R}_i^{-1}\bar{C}_{i,t})^{-1}\bar{C}_{i,t}^\mathsf{T}\bar{R}_i^{-1}, \qquad t \in \{0,\dots,N-k\}, \tag{5}$$

the Gauss–Markov estimator for the model that neglects the drift of θ_i inside the window. Since $\theta_i(t+j) = \theta_i(t) + (A^j - I)\theta_i(t) + \sum_{l<j} A^{j-1-l}w_i(t+l)$ and $K_{i,t}\bar{C}_{i,t} = I_p$, we have

$$\tilde{\theta}_i(t) = \theta_i(t) + \eta_i(t) + \beta_i(t), \tag{6}$$

with the noise $\eta_i(t) := K_{i,t}\bar{w}_{m,i,t}$, where $\bar{w}_{m,i,t}$ stacks $w_{m,i}(t),\dots,w_{m,i}(t+k-1)$, and the bias $\beta_i(t) := K_{i,t}\bar{b}_{i,t}$, where $\bar{b}_{i,t}$ stacks $C_{i,t+j}(\theta_i(t+j) - \theta_i(t))$, $j = 0,\dots,k-1$. The noise is Gaussian and independent of $\eta_i(s)$ for $|t-s| \ge k$, the bias is zero mean and independent of the noise, and across all agents.

Layer 2: Cooperative identification: Substituting (6) into (2) gives, for every agent:

$$\tilde{\theta}_i(t+1) = A\tilde{\theta}_i(t) + \xi_i(t), \qquad \xi_i(t) := \underbrace{w_i(t)}_{\text{process noise}} + \underbrace{\eta_i(t+1) - A\eta_i(t)}_{\text{noise residual}} + \underbrace{\beta_i(t+1) - A\beta_i(t)}_{\text{bias residual}}. \tag{7}$$

Using least squares on (7) is inconsistent, since regressor and residual share $\eta_i(t)$ and $\mathbb{E}[\xi_i(t)\tilde{\theta}_i(t)^\mathsf{T}] \ne 0$ (it contains the term $-A\Sigma_{\eta,i}(t)$). We therefore use the instrument $z_i(t) := \tilde{\theta}_i(t-k)$, and the local moments are:

$$\hat{M}_{j,i} := \tfrac{1}{N-2k}\textstyle\sum_{t=k}^{N-k-1}\tilde{\theta}_i(t+j)z_i(t)^\mathsf{T}, \qquad j \in \{0,1\}. \tag{8}$$

Let $M_{j,i} := \mathbb{E}[\hat{M}_{j,i}]$. Since $w_i(t)$ and $\eta_i(t+1) - A\eta_i(t)$ are independent of $z_i(t)$, $M_{1,i} = AM_{0,i} + \Xi_i$ with $\Xi_i := \frac{1}{N-2k}\sum_t \mathbb{E}[(\beta_i(t+1) - A\beta_i(t))z_i(t)^\mathsf{T}]$, for every i with the *same* A; hence the pooled moments $\bar{M}_j := \frac{1}{m}\sum_i M_{j,i}$ satisfy $\bar{M}_1 = A\bar{M}_0 + \bar{\Xi}$, and $\bar{M}_1\bar{M}_0^{-1}$ identifies

Algorithm 1: Cooperative identification and-tracking

Input: f_i^0, g_i, R_i, $\{x_i^{\mathrm{e}}(t)\}_{t<N}$, k, τ, α, $x_i(0)$, $\gamma_i(0) = y_i(x_i(0), 0)$, $s_i(0) = \gamma_i(0)$
for $t = 0, 1, 2, \dots$ **do**
  if $t < N$ **then** query the oracle at $x_i^{\mathrm{e}}(t)$; store $\tilde{y}_i(t)$, $C_{i,t}$;
  exchange $x_i(t) - \alpha s_i(t)$ with $\mathcal{N}_i$; update $x_i(t+1)$ by (11);
  if $t+1 < N$ **then** $\gamma_i(t+1) = y_i(x_i(t+1), t+1)$ `// oracle gradient`;
  if $t+1 = N$ **then**
    compute $\{\tilde{\theta}_i(t')\}_{t' \le N-k}$ by (5) and $\hat{M}_{0,i}, \hat{M}_{1,i}$ by (8);
    run τ consensus rounds and set $\hat{A}^{(i)}$ by (9)
  end
  if $t+1 \ge N$ **then** forecast $\hat{\theta}_i(t+1)$ by (10); $\gamma_i(t+1) = \hat{y}_i(x_i(t+1), t+1)$ `// forecast gradient`;
  exchange $s_i(t) + \gamma_i(t+1) - \gamma_i(t)$ with $\mathcal{N}_i$;
  update $s_i(t+1)$ by (11);
end

A up to the bias residual $\bar{\Xi}\bar{M}_0^{-1}$. The agents compute the pooled sample moments by τ rounds of average consensus and estimate the dynamics as:

$$\hat{M}_j^{(i)}(r+1) = \textstyle\sum_{l \in \mathcal{N}_i \cup \{i\}} W_{il} \hat{M}_j^{(l)}(r), \qquad \hat{A}^{(i)} := \hat{M}_1^{(i)}(\tau)\big(\hat{M}_0^{(i)}(\tau)\big)^{-1}, \tag{9}$$

for $r = 0, \dots, \tau - 1$, with $\hat{M}_j^{(i)}(0) = \hat{M}_{j,i}$.

Layer 3: Network optimization: For $t \ge N$, agent i propagates its last window estimate through the identified dynamics and forms the forecast gradient

$$\hat{\theta}_i(t) := (\hat{A}^{(i)})^{t-(N-k)} \tilde{\theta}_i(N-k), \qquad \hat{y}_i(x,t) := \nabla_x f_i^0(x) + C_i(x)\hat{\theta}_i(t). \tag{10}$$

The predicted minimizer is $\hat{x}^*(t)$. Every agent maintains a decision $x_i(t)$ and a gradient tracker $s_i(t)$ updated by the adapt-then-combine recursion (Xu et al., 2015; Nedić et al., 2017), with sums over $j \in \mathcal{N}_i \cup \{i\}$ as

$$x_i(t+1) = \textstyle\sum_j W_{ij}\big(x_j(t) - \alpha s_j(t)\big), \qquad s_i(t+1) = \textstyle\sum_j W_{ij}\big(s_j(t) + \gamma_j(t+1) - \gamma_j(t)\big), \tag{11}$$

with step $\alpha > 0$, $s_i(0) = \gamma_i(0)$, and gradient signal $\gamma_i(t)$ equal to the oracle gradient for $t < N$ and to the forecast gradient $\hat{y}_i(x_i(t), t)$ for $t \ge N$.

4. Main results

For $t \ge N$ the error of agent i splits as

$$\|x_i(t) - x^*(t)\| \le \underbrace{\|x_i(t) - \bar{x}(t)\|}_{\text{disagreement}} + \underbrace{\|\bar{x}(t) - \hat{x}^*(t)\|}_{\text{optimization}} + \underbrace{\|\hat{x}^*(t) - x^*(t)\|}_{\text{prediction}}, \tag{12}$$

where $\bar{x}(t) := \frac{1}{m}\sum_i x_i(t)$. By (4) the prediction error is at most $L_* \sum_i \|\hat{\theta}_i(t) - \theta_i(t)\|$, and, with $H := t-(N-k)$, $\tilde{e}_i := \tilde{\theta}_i(N-k) - \theta_i(N-k)$, and $\theta_i(t) = A^H \theta_i(N-k) + \sum_{j<H} A^j w_i(t-1-j)$, the forecast error of agent i decomposes as

$$\hat{\theta}_i(t) - \theta_i(t) = \underbrace{\big((\hat{A}^{(i)})^H - A^H\big)\tilde{\theta}_i(N-k)}_{\text{identification}} + \underbrace{A^H \tilde{e}_i}_{\text{reconstruction}} - \underbrace{\textstyle\sum_{j<H} A^j w_i(t-1-j)}_{\text{process noise}}. \tag{13}$$

All proofs are in the supplementary material.

4.1. Reconstruction and cooperative identification

The estimate $\tilde{\theta}_i(t)$ in (5) is used in the forecast and has reconstruction error, $\mathbb{E}\|\tilde{\theta}_i(t)-\theta_i(t)\|_2$ that depends on the noise and bias in (6).

Lemma 1 (Reconstruction error) *Let Assumptions 2–3 hold,* $\Gamma_i := \sup_{x\in\mathbb{R}^n}\|C_i(x)\|_2 < \infty$, $B_{\theta,i} := \sup_t(\mathbb{E}\|\theta_i(t)\|_2^2)^{1/2}$, *and* $k \geq k_0$. *For all* $i \in \mathcal{V}$ *and* $t \in \{0,\dots,N-k\}$, $\bar{C}_{i,t}^{\mathsf{T}}\bar{R}_i^{-1}\bar{C}_{i,t} \succeq \alpha_{k,i}I_p$ *with* $\alpha_{k,i} := \lfloor k/k_0\rfloor\alpha_0/\lambda_{\max}(R_i)$, *the covariance* $\Sigma_{\eta,i}(t)$ *of* $\eta_i(t)$ *satisfies* $\Sigma_{\eta,i}(t) \preceq \alpha_{k,i}^{-1}I_p$, *and, with* $\chi_i^2 := 2k_0\lambda_{\max}(R_i)/(\alpha_0\lambda_{\min}(R_i))$, *then*

$$\left(\mathbb{E}\|\tilde{\theta}_i(t)-\theta_i(t)\|^2\right)^{1/2} \leq \sqrt{p/\alpha_{k,i}} + \Gamma_i C_\rho \chi_i\Big(\tfrac{k}{\sqrt{3}}\|A-I\|B_{\theta,i} + \sqrt{\tfrac{k}{2}\operatorname{tr}(Q_i)}\Big) := \epsilon_i^{\mathrm{w}}. \tag{14}$$

The noise term in (14) decreases as $1/\sqrt{k}$ (since $\alpha_{k,i}$ grows linearly with k), while the bias term, which comes from the motion of θ_i inside the window, grows with k, where k is a fixed design parameter.

We identify the common dynamics A using the sample moments (8), which every agent computes locally and then averages. Writing the moment of agent i as $M_{0,i} = A^k\Sigma_{\theta,i} + \Delta_{0,i}$, where $A^k\Sigma_{\theta,i}$ is the stationary value of $\mathbb{E}[\theta_i(t)\theta_i(t-k)^{\mathsf{T}}]$, the noise- and bias-free part of the correlation between the regressor $\tilde{\theta}_i(t)$ and the lagged instrument $\tilde{\theta}_i(t-k)$, and $\Delta_{0,i}$ collects the terms involving the windowing bias β_i and the transient $P_{i,0}-\Sigma_{\theta,i}$ of the initial covariance. The pooled moment is $\bar{M}_0 = A^k\bar{\Sigma}_\theta + \bar{\Delta}_0$ with $\bar{\Sigma}_\theta := \frac{1}{m}\sum_i\Sigma_{\theta,i}$ and $\bar{\Delta}_0 = \frac{1}{m}\sum_i\Delta_{0,i}$. We define the invertibility margin $\nu_k := \sigma_{\min}(A^k\bar{\Sigma}_\theta) > 0$ of $\bar{M}_0$, and the heterogeneity $D_M := \max_{j\in\{0,1\},l\in\mathcal{V}}\|M_{j,l}-\bar{M}_j\|_2$, which vanishes for identical agents.

The estimate $\hat{A}^{(i)}$ differs from A due to three reasons. First, the moments are computed from finitely many samples (finite sample error). Second, each agent i after τ consensus rounds, estimates the network average only up to a consensus error of order σ^τ. Third, the error due to bias residual also contributes in the overall error and does not vanish with N.

Theorem 2 (Sample complexity of cooperative identification) *Let Assumptions 2–4 hold,* $k \geq k_0$, $\Gamma_i < \infty$ *for all* i, $\delta \in (0,1/2]$, $N \geq N_0(\delta/(4m))$, $\|\bar{\Delta}_0\| \leq \nu_k/2$, *and* $\epsilon^{\mathrm{m}}(N,\tau,\delta) \leq \nu_k/8$, *where*

$$\epsilon^{\mathrm{m}}(N,\tau,\delta) := c\sqrt{\log(4m/\delta)/N}\left(m^{-1/2}+\sigma^\tau\right) + \sqrt{m}\,\sigma^\tau D_M. \tag{15}$$

where $c > 0$ *is the* concentration constant *of the sample moments, i.e., a constant independent of* N, m *and* δ *such that, for every* $j \in \{0,1\}$, *every weight vector* $\omega \in \mathbb{R}^m$, *and every* $\delta' \in (0,1/2]$, *with probability at least* $1-\delta'$ *whenever* $N \geq N_0(\delta')$,

$$\Big\|\textstyle\sum_{l=1}^m \omega_l\big(\hat{M}_{j,l}-M_{j,l}\big)\Big\|_2 \leq c\,\|\omega\|_2\sqrt{\frac{\log(1/\delta')}{N}} \tag{16}$$

(c *and the burn-in* $N_0(\cdot)$ *are given explicitly in Lemma S1 of the supplementary material). Then there is an event* $\mathcal{E}_\delta$ *with* $\mathbb{P}(\mathcal{E}_\delta) \geq 1-\delta$ *on which, for all* $i \in \mathcal{V}$ *simultaneously,*

$$\|\hat{A}^{(i)}-A\| \leq \underbrace{C_A(1+\sqrt{m}\sigma^\tau)\sqrt{\frac{\log(4m/\delta)}{mN}}}_{\textit{finite-sample error}} + \underbrace{C_W\sqrt{m}\sigma^\tau D_M}_{\textit{heterogeneity}} + \underbrace{C_B(k\|A-I\|_2 + \sqrt{k\operatorname{tr}(\bar{Q})})}_{\textit{bias residual}} =: \epsilon^{id}(\delta), \tag{17}$$

holds, where $\bar{Q} := \frac{1}{m}\sum_i Q_i$ *and the constants* C_A, C_B, C_W *are independent of* N, τ *and* δ.

4.2. Tracking

For $t \geq N$ the oracle is no longer available and agent i replaces its gradient by the forecast gradient $\hat{y}_i(x,t)$ of (10), which is the gradient of the *forecast cost* $\hat{f}_i(x,t) := f_i^0(x) + g_i(x)^\mathsf{T}\hat{\theta}_i(t)$. The algorithm tracks the minimizer $\hat{x}^*(t)$ of $\sum_i \hat{f}_i(\cdot,t)$. Let $x(t)$, $s(t)$, and $\hat{\theta}(t)$ stack the iterates $x_i(t)$, the trackers $s_i(t)$, and the forecasts $\hat{\theta}_i(t)$ of all agents, and let $\bar{s}(t) := \frac{1}{m}\sum_i s_i(t)$. We define:

$$\begin{aligned} a_t &:= \|x(t) - \mathbf{1}_m \otimes \bar{x}(t)\|_2 \quad \text{(iterate disagreement)}, \\ b_t &:= \sqrt{m}\,\|\bar{x}(t) - \hat{x}^*(t)\|_2 \quad \text{(optimization error)}, \\ c_t &:= \|s(t) - \mathbf{1}_m \otimes \bar{s}(t)\|_2 \quad \text{(tracker disagreement)}, \\ d_t &:= \|\hat{\theta}(t+1) - \hat{\theta}(t)\|_2 \quad \text{(forecast drift)}, \end{aligned} \tag{18}$$

with $e_t := [a_t \;\; b_t \;\; c_t]^\mathsf{T}$. The first two are the disagreement and optimization terms of (12), c_t measures how well the trackers follow the average gradient, and d_t bounds the motion of the predicted minimizer, since $\|\hat{x}^*(t+1) - \hat{x}^*(t)\| \leq L_* d_t$ by (4).

Lemma 3 *Let Assumptions 1 and 4 hold, let the forecasts satisfy $\hat{\theta}_i(t) \in \Theta_i$ for all i and $t \geq N$, let $\kappa := mL/\mu$, $\Gamma := \max_i \Gamma_i < \infty$ with Γ_i as in Lemma 1, and let*

$$\alpha \in (0, \alpha_{\max}], \qquad \alpha_{\max} := \min\left\{\frac{(1-\sigma)^2}{4L},\; \frac{1}{(\kappa+1)L}\right\}. \tag{19}$$

Then there exist $\rho_{\mathrm{gt}} \in (0,1)$ and $c_{\mathrm{gt}} > 0$, depending only on $(\sigma, \mu, L, m, \alpha, \Gamma, L_*)$, such that for all $i \in \mathcal{V}$ and $t \geq N$,

$$\|x_i(t) - \hat{x}^*(t)\|_2 \leq c_{\mathrm{gt}}\Big(\rho_{\mathrm{gt}}^{t-N}\|e_N\|_2 + \sum_{q=N}^{t-1} \rho_{\mathrm{gt}}^{t-1-q} d_q\Big). \tag{20}$$

The iterates forget their initial state at the rate ρ_{gt} and lag behind the predicted minimizer by a weighted sum of its past drifts. The drift is produced by the forecasts as the next theorem shows.

The tracking error of agent i thus has three sources. First, the network needs time to reach consensus and to optimize. Second, the predicted minimizer moves while the forecasts are propagated, so the iterates lag behind it. Third, the forecasts are wrong, by the identification, reconstruction, and process-noise terms of (13), which shifts the minimizer away from $x^*(t)$. Since the identification results hold on $\mathcal{E}_\delta$. For a random variable Z, we have $\mathbb{E}_\delta[Z] := \mathbb{E}[Z\mathbb{I}_{\mathcal{E}_\delta}]$ for $Z \geq 0$, so that $\mathbb{E}[Z \mid \mathcal{E}_\delta] \leq \mathbb{E}_\delta[Z]/(1-\delta)$.

Theorem 4 (Distributed tracking error) *Let Assumptions 1–4 and the hypotheses of Theorem 2 and of Lemma 3 hold, and let $\hat{\rho} := \rho_A + C_\rho \epsilon^{\mathrm{id}}(\delta) < 1$ and $\rho_\star := \max\{\rho_{\mathrm{gt}}, \hat{\rho}\}$. Then, for every agent $i \in \mathcal{V}$ and every $t \geq N$,*

$$\mathbb{E}_\delta\|x_i(t) - x^*(t)\| \leq \underbrace{c_{\mathrm{gt}}\rho_{\mathrm{gt}}^{t-N}\mathbb{E}_\delta\|e_N\|}_{\text{(i) optimization error}} + \underbrace{c_{\mathrm{gt}} C_\rho \hat{\rho}^k (t-N)\rho_\star^{t-N-1} U_N}_{\text{(ii) forecast drift}} + \underbrace{L_* \textstyle\sum_{j=1}^m \epsilon_j^{\mathrm{f}}(N,t;\delta)}_{\text{(iii) forecast error}}, \tag{21}$$

Table 1: Parameters of the two case studies. Ranges denote known constants drawn once per instance, uniformly and with a fixed seed; $\mathrm{Rot}(\omega)$ is the planar rotation by ω. In (A) one step is one second and one length unit is 10 m.

	(A) charging station	(B) recommendation
m, n, p	6, 2, 2	8, 3, 3
graph (contraction factor σ)	ring + 3 chords (0.50)	ring + 2 chords (0.68)
shared dynamics A	$0.99\,\mathrm{Rot}(0.15)$	$\mathrm{blkdiag}(0.98\,\mathrm{Rot}(0.25), 0.95)$
Q_i, R_i, $P_{i,0}$	$0.05^2 I$, $1.5^2 I$, $4^2 I$	$0.02^2 I$, $0.15^2 I$, $1.5^2 I$
known constants	$h_i \in [0.5, 1.5]$, $\|\bar{b}_i\| \in [3, 8]$	$\lambda_i \in [0.4, 0.8]$, $\bar{u}_{i,\ell} \in [1, 3]$
exploration $x_i^{\mathrm{e}}(t)$	the iterate $x_i(t)$	0
excitation k_0, α_0	1, $\min_i 4h_i^2 = 3.6$	1, 0.25
window k, step size α	2, 0.2	3, 0.25

where $U_N := \big(\|A - I\| + \epsilon^{\mathrm{id}}(\delta)\big) \sum_{j=1}^{m} \big(B_{\theta,j} + \epsilon_j^{\mathrm{w}}\big)$, $\mathbb{E}_\delta \|e_N\| < \infty$, *and* $\epsilon_j^{\mathrm{f}}(N, t; \delta)$ *bounds the forecast error of agent* j *with* $H = t - (N - k)$ *and* $\sigma_{H,j}^2 := \mathrm{tr} \sum_{l<H} A^l Q_j (A^l)^{\mathsf{T}} \le \mathrm{tr}\, \Sigma_{\theta,j}$,

$$\underbrace{\mathbb{E}_\delta \|\hat{\theta}_j(t) - \theta_j(t)\|}_{\textit{forecast error}} \le \epsilon_j^{\mathrm{f}}(N, t; \delta) := \underbrace{C_\rho^2 H \hat{\rho}^{H-1} \big(B_{\theta,j} + \epsilon_j^{\mathrm{w}}\big)\, \epsilon^{\mathrm{id}}(\delta)}_{\textit{identification}} + \underbrace{C_{\rho\rho} \rho_A^H\, \epsilon_j^{\mathrm{w}}}_{\textit{reconstruction}} + \underbrace{\sigma_{H,j}}_{\textit{process noise}} . \tag{22}$$

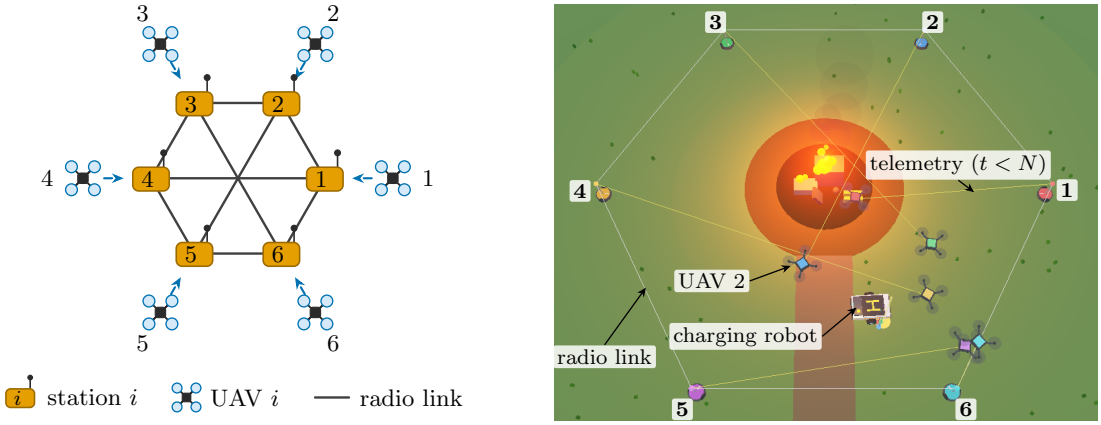
Term (i), the price of distributing the computation, decays geometrically at a rate independent of N, k, and the identification accuracy; term (ii), induced by the motion of the predicted minimizer, vanishes because the forecasts converge; term (iii) is the forecast floor: its identification part is proportional to $\sqrt{\log(4m/\delta)/(mN)}$ plus the bias residual and vanishes with H because the true and the identified dynamics are stable, while its process-noise part saturates at $\sqrt{\mathrm{tr}\, \Sigma_{\theta,j}}$. As $N, H \to \infty$ every agent's error converges to $L_* \sum_j \sqrt{\mathrm{tr}\, \Sigma_{\theta,j}}$, the irreducible error of forecasting an unmeasured stochastic parameter. Cooperation enters only through $\epsilon^{\mathrm{id}}(\delta)$, so its gain is largest when data are scarce.

5. Numerical experiments

We test the predictions of Section 4 on two case studies: the ground stations of a disaster-response UAV fleet that must agree on the position of a shared mobile charging cart (physical agents, quadratic costs) and a decentralized federated recommendation problem (data-holding agents, nonquadratic costs). Section 5.1 presents each case study, its setting followed by its results, and Section 5.2 runs case study (A) as a distributed program in the robot simulator Webots. The Python code that generates every figure, the simulation, and its data collection accompany the paper.

5.1. Case studies and results

(A) A charging cart for patrol UAVs. Figure 1 illustrates the setting and the Webot simulation setup. At a wildfire with search and rescue, six agencies (police, fire department, emergency medical services, search and rescue, utilities, and civil protection) each fly one patrol UAV over the site. UAV i holds a planned position $p_i(t)$, known only to its agency, whose ground station receives the position and telemetry of that UAV only. The agencies



(a) Ground stations and radio links (b) Webots, during data collection

Figure 1: Case study (A). (a) Station i receives the telemetry of UAV i only, during data collection ($t < N$), and exchanges messages only with its three neighbors on the radio network, a ring with three chords. (b) The Webots simulation at $t = 50$: each UAV has the color of its station, yellow lines are telemetry, white lines are radio links, and the charging robot drives to the mean of the stations' estimates.

share neither telemetry, nor video, nor planned positions. The cellular network is limited, where each station exchanges messages only with its nearest stations. A ground robot carries a charging station for the UAVs; a UAV that needs power flies to the cart, charges, and returns to its observation location. The stations must agree on where the cart should be, and move it as the UAVs move. The position of UAV i is its planned position plus a deviation,

$$b_i(t) = p_i(t) + \theta_i(t), \qquad \theta_i(t+1) = A\,\theta_i(t) + w_i(t).$$

In this case study the planned position is a fixed observation location, $p_i(t) \equiv \bar{b}_i$ (Table 1). The deviation $\theta_i(t)$ is unknown and wind gusts $w_i(t)$ push the UAV off its planned position. The UAVs have dynamics $A = \begin{bmatrix} 0.979 & -0.148 \\ 0.148 & 0.979 \end{bmatrix}$, which is unknown to the stations. Putting the cart at x costs UAV i the squared distance to it, weighted by the known constant h_i, denoting the rate at which that UAV needs power:

$$f_i(x, \theta_i) = h_i \|x - b_i(t)\|^2 = h_i \|x - p_i(t) - \theta_i\|^2.$$

The team cost is smallest at the weighted centroid $x^\star(t) = \sum_i h_i b_i(t) / \sum_i h_i$, which moves with the UAVs. We consider $f_i^0(x) = h_i \|x - p_i(t)\|^2$ and $g_i(x) = -2h_i(x - p_i(t))$, which give f_i up to the term $h_i \|\theta_i\|^2$, independent of x, so $C_i = -2h_i I_2$ is invertible and Assumption 3 holds with $k_0 = 1$. A noisy gradient of f_i at any x is a noisy position of UAV i, which station i receives once per second during data collection ($t < N$). At time $t = N$, data collection stops and no station receives the position of its UAV, and each must forecast it to keep the cart in place.

The UAVs start on average 50 m off their planned positions, gusts displace them by 0.5 m per second, and the received positions carry errors of 5 to 15 m. Each station talks only to its three neighbors in the network of Figure 1(a), a ring with three chords ($\sigma = 0.50$).

Results of (A). We run our simulation over 50 Monte Carlo trials for a tracking horizon of 200 steps ($T = N + 200$). The iterates start at $x_i(0) = 0$ with $s_i(0) = \gamma_i(0)$. Cooperative

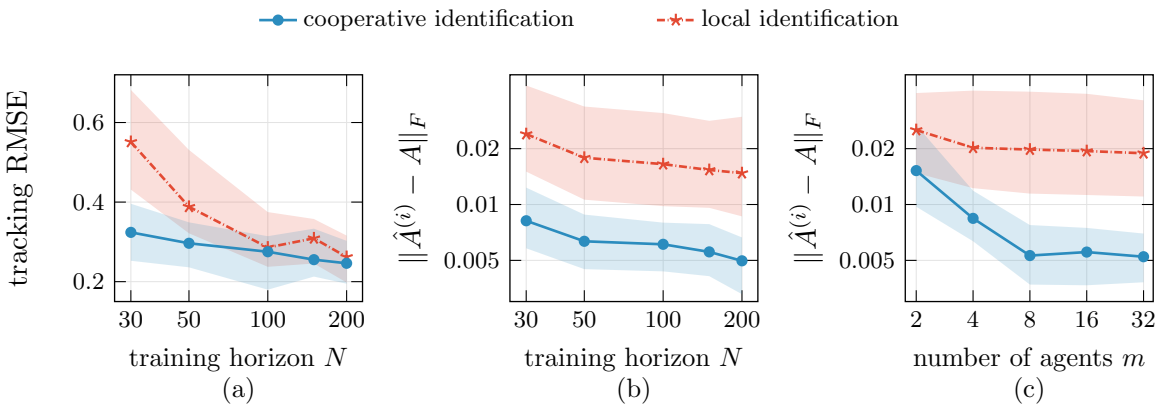


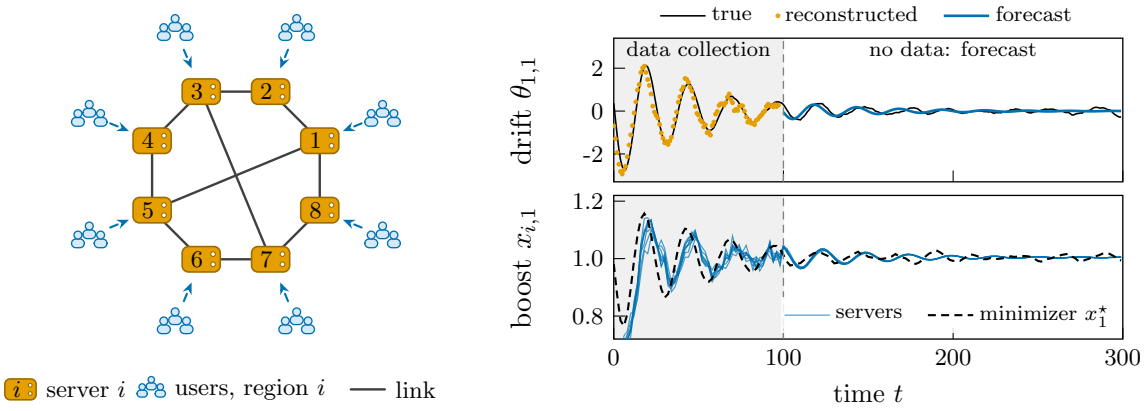
Figure 2: Case study (A), cooperative identification ($\tau = 3$) versus local identification ($\tau = 0$) over 50 trials. Lines are medians and shaded bands are interquartile ranges, taken over the trials for the tracking RMSE and over trials and stations for the identification error. (a) Tracking RMSE (in units of 10 m) versus the training horizon N. (b) Identification error $\|\hat{A}^{(i)} - A\|_F$ versus N. (c) Identification error versus the number m of identical UAVs ($h_i = 1$) on a ring with diametral chords, $N = 100$.

identification has $\tau = 3$ consensus rounds and $\tau = 0$ for local identification. Fig. 2(a) shows the tracking RMSE $\big(\frac{1}{150m}\sum_{t=N+50}^{N+199}\sum_{i=1}^{m}\|x_i(t) - x^*(t)\|^2\big)^{1/2}$ for different values of N. We observe that the tracking error decreases with N. Fig. 2(b) shows the error in identification of dynamics A. We observe that the cooperative system identification is 2.7–3 times more accurate than the local identification by single station for every N. The cooperative error decreases from 0.008 to 0.005 as N grows. Fig. 2(c) shows the identification error with the different number of identical agents, i.e., with $h_i = 1$. We observe that the error for cooperative identification decreases as agents increase (as $1/\sqrt{m}$ from Theorem 2) but saturates for 16 or more agents, where σ reaches 0.85 and 0.96 and σ^τ is no longer small. However, for local identification, the error remains the same.

(B) Decentralized federated recommendation. Figure 3 illustrates the setting. A content platform serves its users from $m = 8$ regional servers. The data of each region stay on its own server, and there is no central aggregator, as in decentralized federated learning (Koloskova et al., 2020). The backbone network is sparse: each server exchanges messages only with two or three peers. The ranking model is shared, so the servers must agree on common boosts $x \in \mathbb{R}^3$, logit shifts that promote or demote three content genres in every feed, and retune them as the interests of the users drift. The interest of the users of region i in genre ℓ is a known average plus a drift,

$$u_{i,\ell}(t) = \bar{u}_{i,\ell} + \theta_{i,\ell}(t), \qquad \theta_i(t+1) = A\,\theta_i(t) + w_i(t).$$

The drift $\theta_i(t)$ is unknown and private to server i, and the noise $w_i(t)$ models short-term fluctuations of taste. The interests of all regions follow the same trend dynamics $A = \mathrm{blkdiag}(0.98\,\mathrm{Rot}(0.25), 0.95)$, which is unknown to the servers: a damped seasonal oscillation shared by the first two genres and a fading interest in the third. Under the boost x_ℓ, a user engages with an item of genre ℓ with probability $\varsigma(x_\ell)$, where $\varsigma(z) = 1/(1+e^{-z})$ is the logistic function, so the negative log-likelihood of an engagement is $-\log\varsigma(x_\ell) = \mathrm{softplus}(-x_\ell)$.



(a) Regional servers and backbone links

(b) One trial, $N = 100$

Figure 3: Case study (B). (a) Server i receives the engagement data of the users of region i only, during data collection ($t < N$), and exchanges only x_i, s_i, and 3×3 moment matrices with its neighbors on the backbone. (b) Top: drift $\theta_{1,1}(t)$ of the interest of region 1 in genre 1, its window reconstruction by server 1 ($t < N$, shaded), and its forecast with the cooperative estimate of A ($t \geq N$). Bottom: boost $x_{i,1}(t)$ of genre 1 at the eight servers and the minimizer $x_1^\star(t)$ of the team cost.

Server i weights this loss by the interest of its users in each genre and adds a ridge penalty with a known weight $\lambda_i > 0$, which keeps the boosts small:

$$f_i(x, \theta_i) = \frac{\lambda_i}{2}\|x\|^2 + \sum_{\ell=1}^{3} u_{i,\ell}(t)\,\mathrm{softplus}(-x_\ell).$$

The more interested the users are in a genre, the more a missed engagement with it costs, and the stronger the boost of that genre. Unlike in (A), the team cost is not quadratic, and its minimizer $x^\star(t)$ has no closed form; it moves with the interests. We consider $f_i^0(x) = \frac{\lambda_i}{2}\|x\|^2 + \sum_\ell \bar{u}_{i,\ell}\,\mathrm{softplus}(-x_\ell)$ and $g_i(x) = \mathrm{softplus}(-x)$ (elementwise), so $C_i(x) = -\operatorname{diag}(\varsigma(-x))$ is invertible for every x and Assumption 3 holds with $k_0 = 1$. A noisy gradient of f_i at the unboosted configuration $x_i^{\mathrm{e}}(t) = 0$ is a noisy measurement of the interests of region i, which server i receives once per step during data collection ($t < N$). At time $t = N$, data collection stops and no server receives gradients, and each must forecast the interests of its users to keep the boosts tuned.

The drifts start with standard deviation 1.5, the noise $w_i(t)$ has standard deviation 0.02, and the gradients carry noise of standard deviation 0.15. Each server talks only to its neighbors in the backbone of Figure 3(a), a ring with the chords $(1, 5)$ and $(3, 7)$, and sends only x_i, s_i, and 3×3 moment matrices, so no user data leave a region.

Results of (B). We run 50 Monte Carlo trials with the same tracking horizon, initialization, and tracking RMSE as in (A). Figure 3(b) shows one trial with $N = 100$: while $t < N$ the servers reconstruct the seasonal oscillation of the interests from their gradients, and after $t = N$ they forecast it with the cooperative estimate of A and keep their boosts on the moving minimizer without receiving any data. Fig. 4(a) shows the tracking RMSE versus N. With cooperative identification it is 24–37% lower than with local identification for every N, and within 3% of an oracle that knows A. Fig. 4(b) shows the error in identification of A.

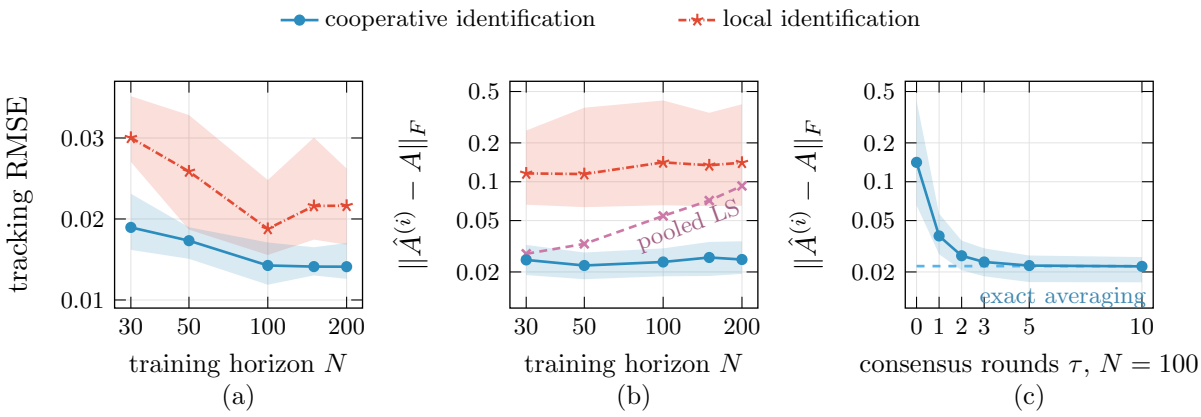


Figure 4: Case study (B), cooperative identification ($\tau = 3$) versus local identification ($\tau = 0$) over 50 trials. Lines are medians and shaded bands are interquartile ranges, as in Figure 2. (a) Tracking RMSE versus the training horizon N. (b) Identification error $\|\hat{A}^{(i)} - A\|_F$ versus N, with pooled least squares (pooled LS). (c) Identification error versus the number τ of consensus rounds, $N = 100$, with the value of exact averaging ($\tau = \infty$).

Cooperation reduces it 4.7–5.9 times, more than $\sqrt{m} = 2.83$, because the local estimator fails outright in many trials: 29–33% of the local estimates are not Schur stable, against none of the cooperative ones. The cooperative error stays between 0.022 and 0.026 for every N. This is the bias floor of Theorem 2, which does not decrease with N and grows with $k\|A - I\|$: $k = 3$ and $\|A - I\| = 0.25$ here, against $k = 2$ and $\|A - I\| = 0.15$ in (A). Pooled least squares, which suffers from the errors-in-variables bias, deteriorates with N instead, from 0.028 to 0.093. Fig. 4(c) varies the number of consensus rounds at $N = 100$ ($\sigma = 0.68$). The identification error decreases from 0.141 for $\tau = 0$ to 0.038 after one round, 0.024 after three, and 0.022 after five, the value of exact averaging, at least as fast as σ^τ predicts. The tracking RMSE drops from 0.019 for $\tau = 0$ to 0.014 for $\tau = 1$ and stays within 2% of its final value for larger τ, so one or two rounds, i.e., $2p^2\tau \leq 36$ reals per link, recover almost all of the cooperative gain. Against the remaining baselines, the tracking RMSE is 2.2–6.3 times lower than with persistence, which tracks with the last window estimate frozen ($\hat{A}^{(i)} = I$), and at most 13% above that of a centralized predictor that receives all forecasts and minimizes $\sum_i \hat{f}_i(\cdot, t)$ exactly (2% for $N \geq 100$), which is the price of the distributed computation.

Summary. In both case studies cooperation reduces the identification error by the factor $\sqrt{m}$ of Theorem 2, or more when local identification is unreliable; the cooperative error settles at the bias floor set by k and $\|A - I\|$, whereas pooled least squares deteriorates with N; a few consensus rounds suffice unless σ is close to one; and the tracking error with cooperative identification is within a few percent of an oracle that knows A as soon as $N \geq 100$, so the gain of cooperation is largest when data are scarce, as Theorem 4 anticipates. In case study (A) this means that six stations that never exchange a telemetry sample keep the cart within about 3 m of its best position for 200 s without seeing their UAVs. What remains is the lag of gradient tracking behind a fast-moving minimizer, term (ii) of Theorem 4, which cooperation cannot remove.

5.2. Simulation in Webots

Setting. To test Algorithm 1 as a distributed program, and not only as a stacked recursion, we implemented case study (A) in the open-source robot simulator Webots (Michel, 2004) (Figure 1(b)). Every ground station is a separate controller process that holds only its own telemetry and its own variables x_i, s_i, $\hat{M}_{0,i}$, $\hat{M}_{1,i}$; the stations interact only through simulated radio emitters and receivers, and each uses only the packets of its two nearest neighbors, a ring with $\sigma = 2/3$, sparser than the graph of Table 1. A supervisor process plays the environment: it propagates $\theta_i(t)$ by (2) and sends to station i alone a noisy position of UAV i while $t < N$; it relays no message between stations and does not move the charging station. The charging station is a wheeled ground robot, simulated with its wheel–ground contact, whose own controller receives the stations' iterates by radio and drives to their mean; UAVs land on its pad to recharge. One step of Algorithm 1 spans ten simulator steps of 32 ms: each of the two exchanges of (11) takes one, and the three consensus rounds at $t = N - 1$ take three more, and the service rates are fixed values in the range of Table 1. Otherwise the instance is case study (A) of Section 5.1, with three changes chosen to make the charging station move and to keep the scene readable. The UAVs fly a known search pattern instead of holding their observation locations: they keep a line abreast, 90 m long, whose centre circles the site at a radius of 90 m once every 600 s, and each flies a small loop in its own lane, so that $x^\star(t)$ travels around the site, about 550 m per lap, with the routes as well as with the deviations, and the known part of the cost, $f_i^0(x) = h_i\|x - p_i(t)\|^2$, varies with time; Algorithm 1 uses it unchanged, but our analysis assumes a static f_i^0. The initial deviation is smaller, $P_{i,0} = 2.5^2 I$. And every telemetry position has an error of standard deviation 0.75, so that $R_i = (1.5h_i)^2 I$, which equals the R_i of Table 1 for $h_i = 1$.

Data collection. For each $N \in \{30, 50, 100, 150, 200\}$ we ran 20 seeds with a tracking horizon of 200 steps. Every run records, at every step, the telemetry that each station received, its iterate, its forecast, its estimate $\hat{A}^{(i)}$, and every packet it sent. Applied offline to the recorded telemetry, the reference implementation behind Figures 2–4 reproduces the stations' estimates to within $6 \cdot 10^{-16}$, and replaying (11) reproduces their iterates to within $8 \cdot 10^{-15}$: the runs are exact executions of Algorithm 1. Local identification, exact averaging, the oracle, and persistence are replayed on the same telemetry.

Results. Over all 100 runs, the simulation reproduces the findings of Section 5.1 on a sparser graph, with moving routes, and with a charging station that has to drive. Cooperation reduces the median identification error from 0.030–0.042 to 0.011–0.014, by a factor of 2.6–3.1 for every N, at least $\sqrt{m} = 2.45$, and three rounds stay within 14% of exact averaging although $\sigma^3 = 0.30$; 23–32% of the local estimates are not Schur stable before the safeguard (an estimate with spectral radius above $\rho_{\max} = 0.999$ is rescaled to that radius; all estimators use it), against at most 4% of the cooperative ones, and none for $N \geq 150$. The errors level off for $N \geq 50$ and are about twice those of Figure 2(b) because the smaller initial deviation excites the dynamics less (a synthetic check with the reference code gives a median error of 0.0125 for $P_{i,0} = 2.5^2 I$ against 0.0066 for the $4^2 I$ of Table 1; $N = 100$, exact averaging). In tracking, the median RMSE is 16–39% lower with cooperative than with local identification, within 16% of the oracle that knows A (3% for $N \geq 100$), and 1.4–4.0 times lower than with persistence. With $N = 100$ the iterates stay within 3.8 m (RMS) of $x^\star(t)$ through the outage while it travels around the site, against 4.5 m with local identification and 8.3 m

with persistence, and the charging robot, which drives after their mean, within 2.9 m. Each station sends 4 reals per step and 24 for the identification, 1222 reals in 602 packets in a run with $N = 100$ (46 kB in the simulation's JSON encoding), and never a position; it also sends its iterate to the robot. Such a run takes 4.3 s of wall-clock time on a laptop.

6. Conclusion

We studied online optimization over a network of agents whose local costs depend on private latent parameters with shared unknown stochastic dynamics, observed only through finitely many noisy gradients. Averaging instrumental-variable moments over the network learns the dynamics with the sample complexity of the pooled data set without exchanging data, and gradient tracking on the forecast gradients tracks the minimizer of the forecast costs up to a network-dependent transient. Future directions include coupled dynamics across agents and time-varying graphs.

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